

Gaoxiang Zhao

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Education

University of Pennsylvania

August 2025 – May 2027

MS in Robotics

Philadelphia, USA

- Related Courses: Machine Learning (A), Learning in Robotics (A), Physical Intelligence (A)

Wuhan University

September 2021 – July 2025

BEng in Communication Engineering

Wuhan, China

Experience

University of Pennsylvania

December 2025 – Present

Research Intern (Advisor: Kostas Daniilidis)

Philadelphia, USA

- Working on event-based vision and its applications in robotics.

Manycore Tech

August 2024 – November 2024

Research Intern

Hangzhou, China

- Developed Monte-Carlo physically-based rendering algorithms for a production-scale rendering pipeline supporting photorealistic synthetic data generation.

Projects

Sim2Real Drone Racing with Deep RL (ESE 6510)

February 2026 – April 2026

- Designed a PPO-based RL pipeline in Isaac Lab with 8,192 parallel environments to train a Crazyflie 2.1 for autonomous 3-lap, 7-gate racing, using a 36D body-frame observation (velocity, rotation matrix, gate corners) and a 10-term reward function.
- Achieved zero-shot sim-to-real transfer via dynamics domain randomization (thrust-to-weight, drag, PID gains) and three iteration-conditioned curricula (approach distance, DR magnitude, speed reward weighting), completing 3 laps in 20s on a real Crazyflie 2.1.

Forward and Inverse Kinematics Solver (CIS 5620)

September 2025 – November 2025

- Developed a C++ kinematics toolkit with forward/inverse kinematics and quaternion-based orientation interpolation.
- Implemented three IK solvers (Limb-based, CCD, Pseudo-Inverse Jacobian).

Physically-based Renderer (Personal Project)

December 2023 – January 2025

- Developed a physically-based rendering engine in C++ with support for diverse geometric primitives, materials, textures, and light sources.
- Implemented path tracing, bidirectional path tracing, and photon mapping for light transport simulation in complex scenes.

Diffraction Simulation

June 2024 – August 2024

- Implemented the Free-Space Diffraction BSDF (Steinberg et al., 2024) in the PBRT-v3 renderer, extending Monte Carlo path tracing with edge-based Fraunhofer diffraction.
- Added continuous spectral rendering on top of the original monochromatic and RGB tricolor modes to study wavelength discretization effects on diffraction patterns.

Technical Skills

- Programming: C++, Python, MATLAB
- Tools: $\LaTeX$, Linux, Git, Slurm, PyTorch, ROS, Unity, OpenGL, Isaac Lab, MuJoCo
- Language: English (Proficient), Mandarin (Native)